VIT Vidyalankar Institute of Technology
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Project Report

by TE EXTC students

Health Monitoring System Using STM32 Blue Pill

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A report submitted in partial fulfilment of the degree of

TE in EXTC

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Declaration

We, hereby declare that our project "**Health Monitoring System Using STM32 Blue Pill "** is submitted by us towards MiniProject of Semester 5 i.e. Third Year of Engineering of B.E. Degree, in the Department of Electronics and Telecommunication Engineering. A project work carried by us under the supervision of Mr. Vijay Purohit, and have not been submitted anywhere else.

We will be solely responsible if any kind of plagiarism is found.

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CERTIFICATE

This is to certify that the project entitled **"Health Monitoring System Using STM32 Blue Pill"** has been submitted to the Department of EXTC Engineering, of Vidyalankar Institute of Technology, University of Mumbai for the fulfilment of the requirement for the award of the Bachelor Degree of Technology in "Electronics and Telecommunication Engineering" by following Student of Third year B.E. (EXTC Engineering).

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Acknowledgement

We like to share our sincere gratitude to all those who help us in completion of this project. During the work we faced many challenges due to our lack of knowledge and experience but these people helped us to get over all the difficulties and in final compilation of our idea to a shaped sculpture.

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We are also thankful to our whole class and most of all to our parents who have inspired us during this pandemic to face all the challenges and overcome all the hurdles in life.

Thank you All,

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Abstract

Health Monitoring System i.e. a virtual medical treatment has become very common during this Covid-19 pandemic to have a check on chronic diseases. Our Health Monitoring System checks temperature of the body and oxygen level with the help of microcontroller STM-32F103C8T6 i.e. Blue Pill. We have used various sensors to achieve this.To check body temperature we have used an IC named LM35, to check oxygen level i.e. oximeter we have used a sensor named MAX30100. Further to display the result of temperature and oximeter sensor we have used a WIFI module named ESP8266 .With the help of WIFI module result would be displayed on ThinkSpeak an IOT platform.

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CHAPTER 1:

INTRODUCTION

Health is always a major concern for every human being. Like the recent coronavirus attack that has ruined the economy of China to an extent is an example of how health care has become of major importance. In such areas where the epidemic is spreading, it is always a better idea to monitor these patients using remote health monitoring technology. Since IoT technology is becoming more and more popular in the commercial market. So we have made our project IOT based only i.e.The Health Monitoring System which is the current solution for it. Remote Patient Monitoring arrangement empowers observation of patients outside of customary clinical settings (e.g. at home), which expands access to human services offices to bring down expenses. In most of the rural areas, the medical facility would not be within a hand reach distance for the natives. So normally people neglect any kind of minor health issues which are shown in early stages by variation of vital elements like body temperature, heartbeat rate etc. Once the health issue has been increased to a critical stage and the life of the person is endangered.

The core objective of this project is the design and implementation of a Health Monitoring System that uses Sensors to track Patient health to inform the patient in case of any issues. The objective of developing health monitoring systems is to reduce health care costs by reducing physician clinic visits, hospitalizations, and diagnostic testing procedures. With tons of new healthcare technology start-ups, IoT is rapidly revolutionizing the healthcare industry. In this project, we have designed the IoT Based Health Monitoring System using l.

CHAPTER 2:

Literature Survey

2.1 Selection of Project

Selection for a project is it a small or a large one is difficult since we have to keep in mind several things like online simulation, availability of components in the market, the overall cost of the project, etc. Keeping all such parameters in mind we have searched a lot for our project. We have shortlisted 2 -3 projects after having a word with our mentor we finalize our Mini Project Topic as Health Monitoring System using STM32 -Blue Pill and WIFI Module.

2.2 Circuit Diagram

Block Diagram

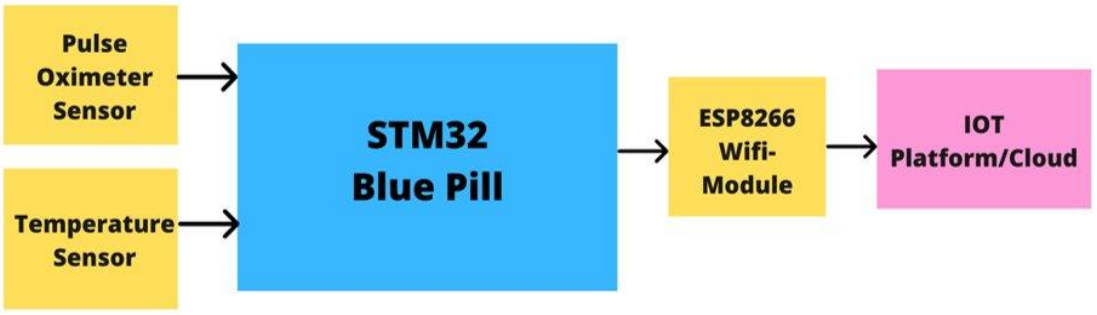


Figure 1. Block Diagram

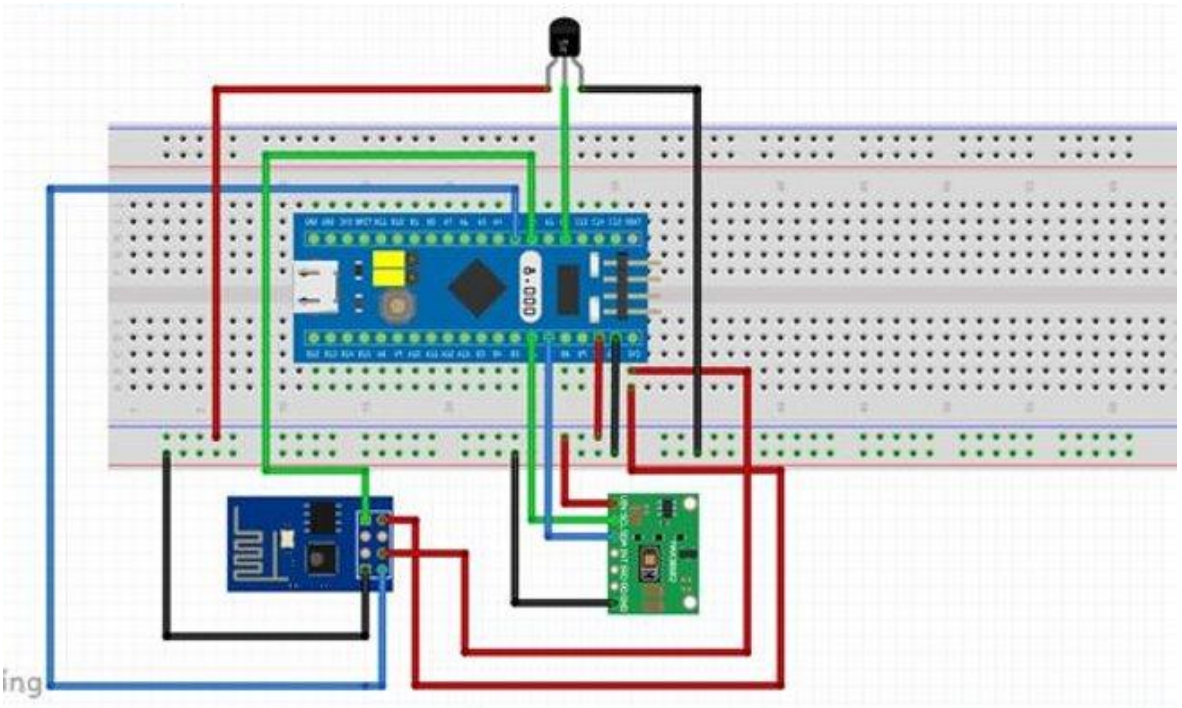
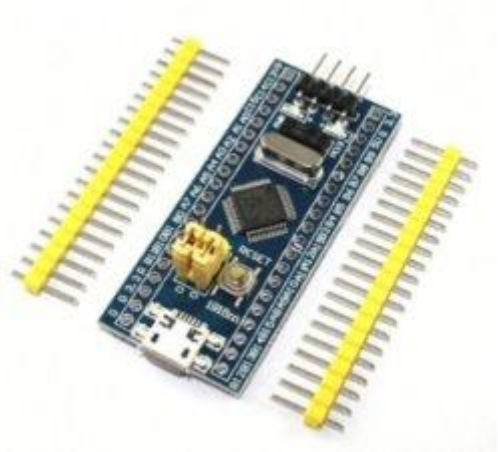


Figure 2. Circuit Diagram

CHAPTER 3:

COMPONENTS REQUIRED



STM32



MAX30100



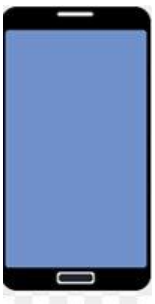
ESP8266



LM35



Jumper Wires



IoT Platform

CHAPTER 4:

SPECIFICATIONS OF COMPONENTS

4.1 Stm32BluePill

STM32F103C8T6 Technical Specifications

Microcontroller	STM32F103C8T6
Operating voltage	3.3V
Analog inputs	10
Digital I/O pins	37
Flash memory (KB)	64/128
SRAM	20KB
Frequency (clock speed)	72MHz max.
Communication	I2C, SPI, UART, CAN, USB

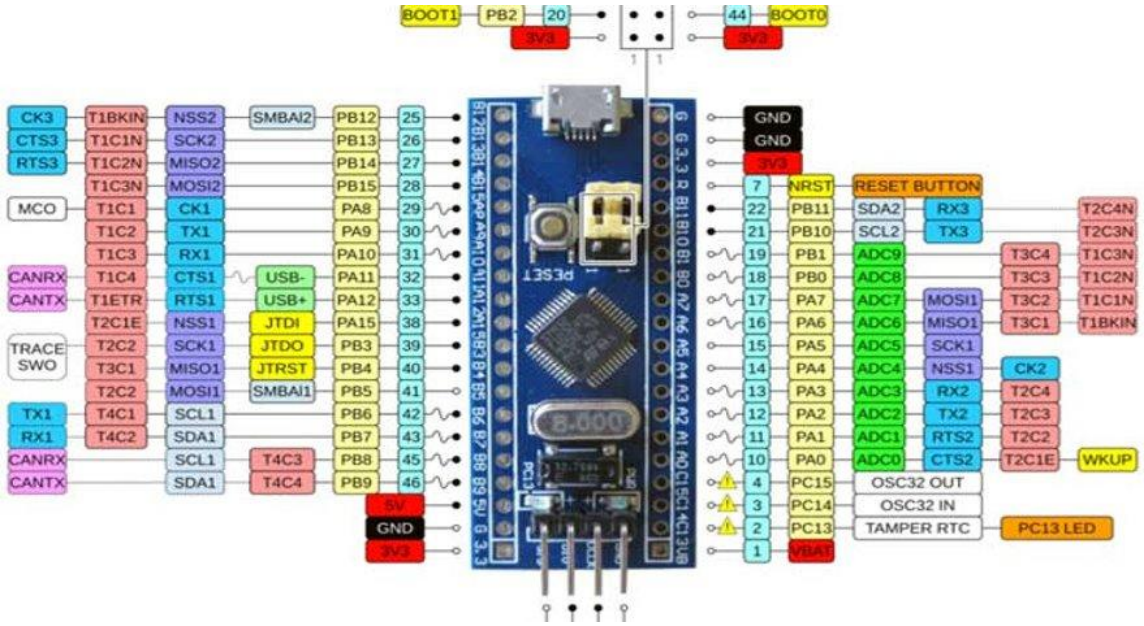


Figure 3. Pin Diagram of STM32 Blue Pill

5

STM32F103C8T6 Blue Pill Pinout Configuration

Category	Pin Name	Details
Power	3.3V, 5V, GND	1. 3.3V – Regulated output voltage from the onboard regulator (drawing current is not recommended), can also be used to supply the chip. 2. 5V from USB or onboard regulator can be used to supply the onboard 3.3V regulator. 3. GND – Ground pins
Analog Pins	PA0 – PA7 PB0 – PB1	Pins act as ADCs with 12-bit resolution
Input/output pins	PA0 – PA15 PB0 – PB15 PC13 – PC15	37 General-purpose I/O pins.
Serial	TX1, RX1 TX2, RX2 TX3, RX3	UART with RTS and CTS pins
External interrupts	PA0 – PA15 PB0 – PB15 PC13 – PC15	All digital pins have interrupt capability
PWM	PA0 – PA3 PA6 – PA10 PB0 - PB1 PB6 – PB9	15 PWM pins total
SPI	MISO0, MOSI0, SCK0, CS0 MISO1, MOSI1, SCK1, CS0	2 SPI
Inbuilt LED	PC13	LED to act as a general-purpose GPIO indicator
I2C	SCL1, SDA1 SCL2, SDA2	Inter-Integrated Circuit communication ports
CAN	CAN0TX, CAN0RX	CAN bus ports

4.2 MAX30100

Pulse Oximeter Sensor

- Pulse Oximeters are low-cost non-Invasive medical sensors used to continuously measure haemoglobin's oxygen saturation (SPO2) in the blood. It displays the percentage of blood that is loaded with oxygen.
- To calculate Blood Oxygen Concentration and Pulse rate we will be using MAX30100 Pulse Oximeter Sensor Module
- The MAX30100 is an integrated pulse oximeter and heart rate monitor sensor solution. It combines two LEDs, a
- the photodetector, optimized optics, and low-noise analog signal processing to detect pulse oximetry and heart-rate signals.



Figure 4. Pulse Oximeter Sensor MAX30100

Features of MAX30100

- Operating Voltage - 1.8V to 5.5V
- Input Current - 20mA
- Integrated Ambient Light Cancellation
- High Sample Rate Capability
- Fast Data Output Capability

Pin Configuration

Pin No.	Signal Name	Description
1	VIN	1.8V – 5.5V Power Input
2	SCL	I2C Serial Clock
3	SDA	I2C Serial Data
4	INT	MAX30100 Interrupt
5	IRD	IR LED Cathode and LED Driver Connection Point. Leave floating in the circuit.
6	RD	Red LED Cathode and LED Driver Connection Point. Leave floating in the circuit.
7	GND	0V / Reference Voltage

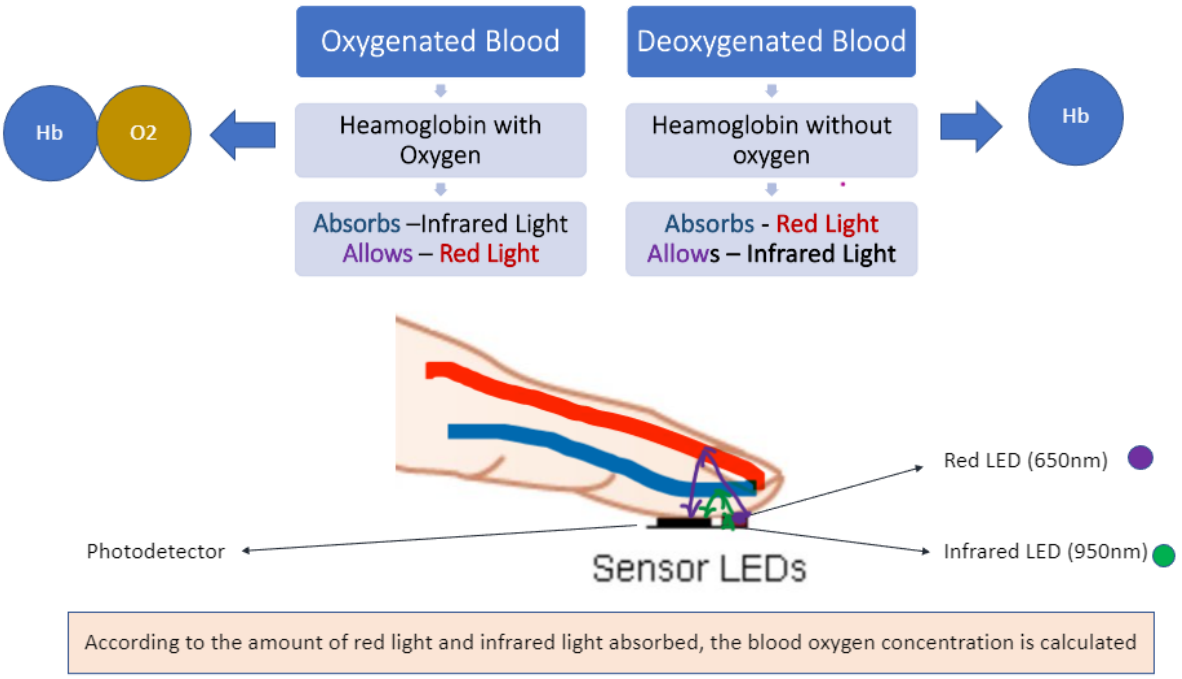


Figure 5. Pulse Oximeter Sensor MAX3010 working

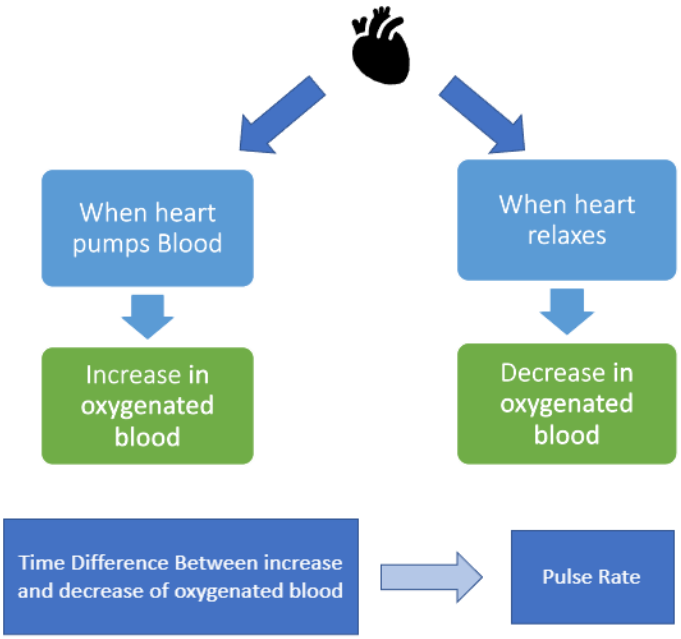


Figure 6. Pulse Oximeter Sensor MAX3010 working block diagram

4.3 ESP8266

ESP8266 Wi-fi Module Features

- Low cost, compact and powerful Wi-Fi Module
- Power Supply: +3.3V only
- Current Consumption: 100mA
- I/O Voltage: 3.6V (max)
- I/O source current: 12mA (max)
- Built-in low power 32-bit MCU @ 80MHz
- 512kB Flash Memory
- Can be used as Station or Access Point or both combined
- Supports Deep sleep (<10uA)
- Supports serial communication (UART) hence compatible with many development platforms like Arduino
- Can be programmed using Arduino IDE or AT-commands or Lua Script

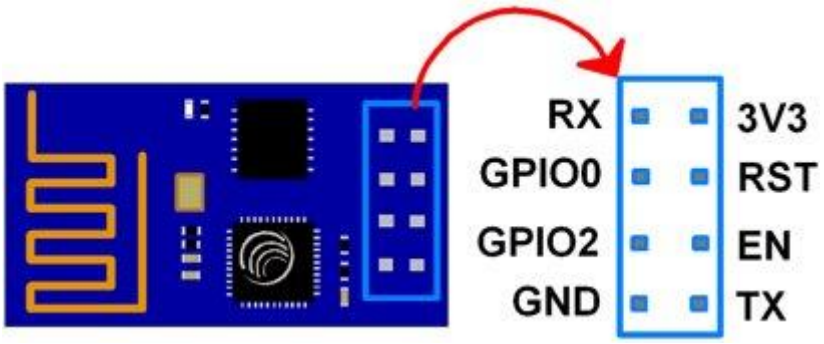


Figure 7. ESP8266 Wifi Module

ESP8266 Wi-fi Module Pin Configuration

Pin Number	Pin Name	Alternate Name	Normally used for	Alternate purpose
1	Ground	-	Connected to the ground of the circuit	-
2	TX	GPIO – 1	Connected to Rx pin of programmer/uC to upload program	Can act as a general-purpose input/output pin when not used as TX
3	GPIO-2	-	General-purpose input/output pin	-
4	CH_EN	-	Chip Enable – Active high	-
5	GPIO - 0	Flash	General-purpose input/output pin	Takes module into serial programming when held low during start-up
6	Reset	-	Resets the module	-
7	RX	GPIO - 3	General-purpose input/output pin	Can act as a general-purpose Input/output pin when not used as RX
8	Vcc	-	Connect to +3.3V only	

Esp 8266 wifi module can be programmed using At commands, serial monitor.

The output thus can be further transmitted via esp to Thingspeak platform

4.4 LM35 Temperature Sensor

- The temperature sensor is a device that is designed to measure temperature.
- LM35 converts body temperature in terms of Analog Voltage
- The LM35 series are precision integrated-circuit temperature devices with an output voltage linearly proportional to the Centigrade temperature.
- The IC LM35 has an advantage over linear temperature sensors calibrated in kelvin.

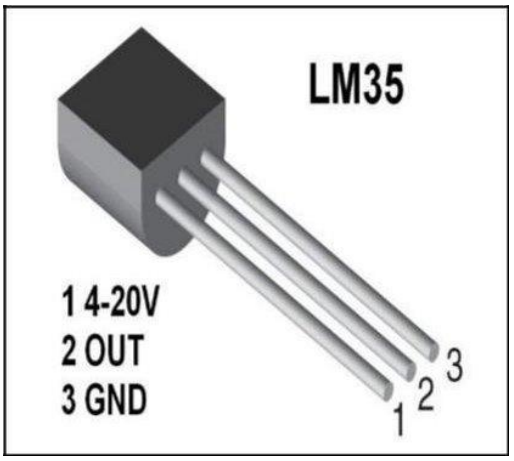


Figure 8. IC LM35

Features of LM35

1. Rated for full −55°C to 150°C Range
2. ±1°C Accuracy over Full Range
3. Operates from 4V to 30V
4. Low Self Heating (60uA)
5. Linear + 10-mV/°C Scale Factor
6. Low Output impedance (0.1 Ω for 1mA load)
7. Suitable for Remote applications.

4.5 IoT Platform

ThingSpeak



Figure 9. Thingspeak platform

ThingSpeak is open-source software written in Ruby which allows users to communicate with internet-enabled devices. It facilitates data access, retrieval, and logging by providing an API to both the devices and social network websites.

ThingSpeak is an IoT analytics platform service that allows you to aggregate, visualize, and analyze live data streams in the cloud. You can send data to ThingSpeak from your devices, create instant visualizations of live data, and send alerts using web services like Twitter® and Twilio®. With MATLAB® analytics inside ThingSpeak, you can write and execute MATLAB code to perform preprocessing, visualizations, and analyses. ThingSpeak enables engineers and scientists to prototype and builds IoT systems without setting up servers or developing web software.

Chapter 5:
Software Used

5.1 Fritzing

Fritzing is an open-source hardware initiative that makes electronics accessible as a creative material for anyone. We offer a software tool, a community website and services in the spirit of Processing and Arduino, fostering a creative ecosystem that allows users to document their prototypes, share them with others, teach electronics in a classroom, and layout and manufacture professional PCBs. The software is created in the spirit of the Processing programming language and the Arduino microcontroller and allows a designer, artist, researcher, or hobbyist to document their Arduino based prototype and create a PCB layout for manufacturing. The associated website helps users share and discuss drafts and experiences as well as reduce manufacturing costs. Fritzing can be seen as an electronic design automation (EDA) tool for non-engineers: the input metaphor is inspired by the environment of designers (the breadboard-based prototype), while the output is focused on accessible means of production. As of December 2, 2014, Fritzing has made a code view option, where one can modify code and upload it directly to an Arduino device. Component images are distributed under CC-BY-SA-3.0, which will also be the license for any generated breadboard views.

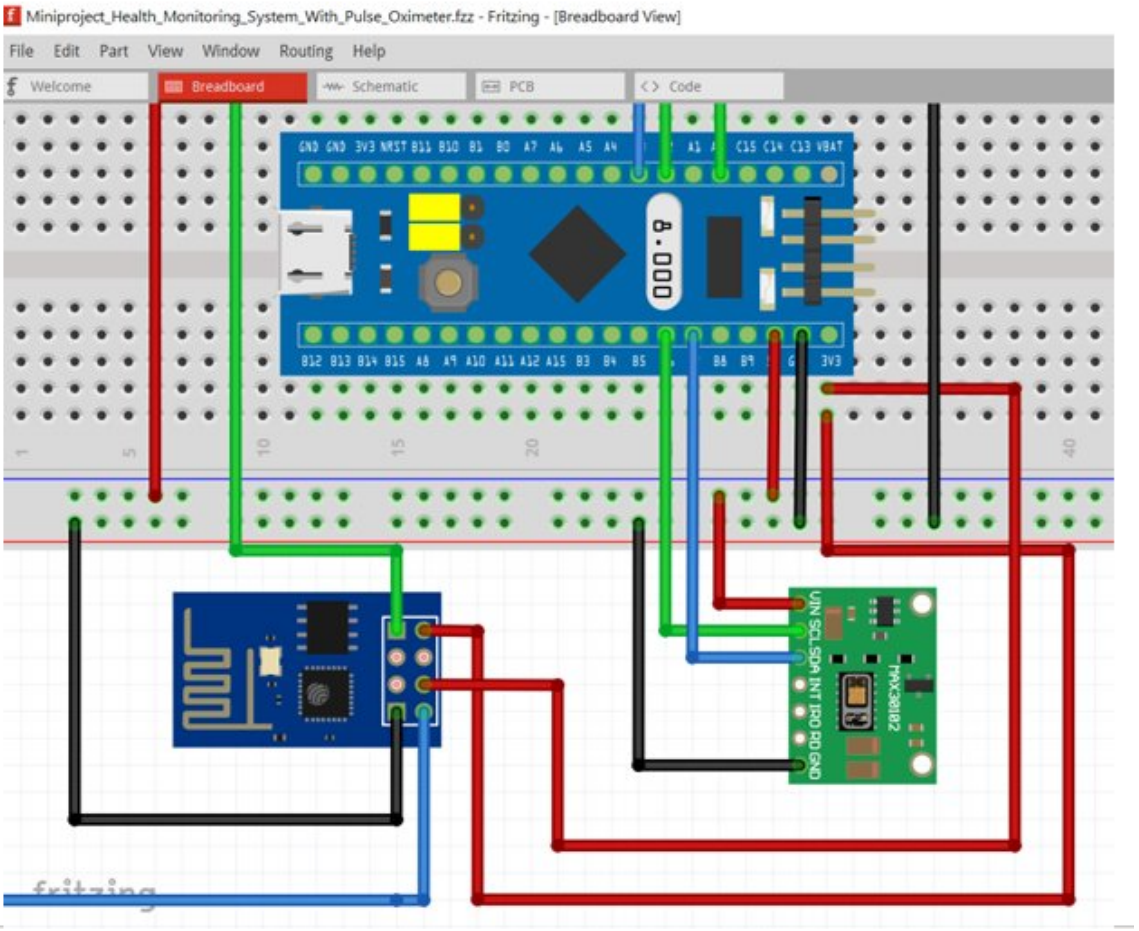


Figure 10. Fritzing

5.2 Arduino IDE

The Arduino Integrated Development Environment (IDE) is a cross-platform application (for Windows, macOS, Linux) that is written in functions from C and C++. It is used to write and upload programs to Arduino compatible boards, but also, with the help of third-party cores, other vendor development boards. The source code for the IDE is released under the GNU General Public License, version 2. The Arduino IDE supports the languages C and C++ using special rules of code structuring. The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures.
User-written code only requires two basic functions, for starting the sketch and the main program loop, that is compiled and linked with a program stub main() into an executable cyclic executive program with the GNU toolchain, also included with the IDE distribution. The Arduino IDE employs the program avrdude to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program in the board's firmware. By default, avrdude is used as the uploading tool to flash the user code onto official Arduino boards.

Arduino IDE is a derivative of the Processing IDE, however, as of version 2.0, the Processing IDE will be replaced with the Visual Studio Code-based Eclipse Theia IDE framework.



Figure 11. Arduino IDE

CHAPTER 6:

CODE

```
#include <Wire.h>
#include "MAX30100_PulseOximeter.h"

#define REPORTING_PERIOD_MS     1000

int val;
int tempPin = PA0;



// PulseOximeter is the higher level interface to the sensor
// it offers:
//  * beat detection reporting
//  * heart rate calculation
//  * SpO2 (oxidation level) calculation
PulseOximeter pox;

uint32_t tsLastReport = 0;

// Callback (registered below) fired when a pulse is detected
void onBeatDetected()
{
    Serial.println("Beat!");
}

void setup()
```

```
{
  Serial.begin(9600);
  while(!Serial){delay(100);}

  Serial.print("Initializing pulse oximeter..");

  // Initialize the PulseOximeter instance
  // Failures are generally due to an improper I2C wiring, missing power supply
  // or wrong target chip
  if (!pox.begin()) {
    Serial.println("FAILED");
    for(;;);
  } else {
    Serial.println("SUCCESS");
  }

  // The default current for the IR LED is 50mA and it could be changed
  //   by uncommenting the following line. Check MAX30100_Registers.h for all the
  //   available options.
  // pox.setIRLedCurrent(MAX30100_LED_CURR_7_6MA);

  // Register a callback for the beat detection
  pox.setOnBeatDetectedCallback(onBeatDetected);
}

void loop()
{
  // Make sure to call update as fast as possible
```

```
pox.update();


// Asynchronously dump heart rate and oxidation levels to the serial
// For both, a value of 0 means "invalid"
if (millis() - tsLastReport > REPORTING_PERIOD_MS) {
   Serial.print("Heart rate:");
   Serial.print(pox.getHeartRate());
   Serial.print("bpm / SpO2:");
   Serial.print(pox.getSpO2());
   Serial.println("%");

   tsLastReport = millis();
   }

   delay(1500);



   val = analogRead(tempPin);
   float t = val * 3300;
   float mv = t / 4096;
   float cel = mv/10;
   float farh = (cel * 9) / 5 + 32;
   Serial.print("TEMPERATURE IN CELSIUS = ");
   Serial.print(cel);
   Serial.print("*C");
   Serial.println();
   delay(1500);
   /* uncomment this to get temperature in fahrenheit */
```

```
        Serial.print("TEMPERATURE IN FAHRENHEIT = ");
        Serial.print(farh);
        Serial.print("*F");
        Serial.println();
        delay(1500);


}
```

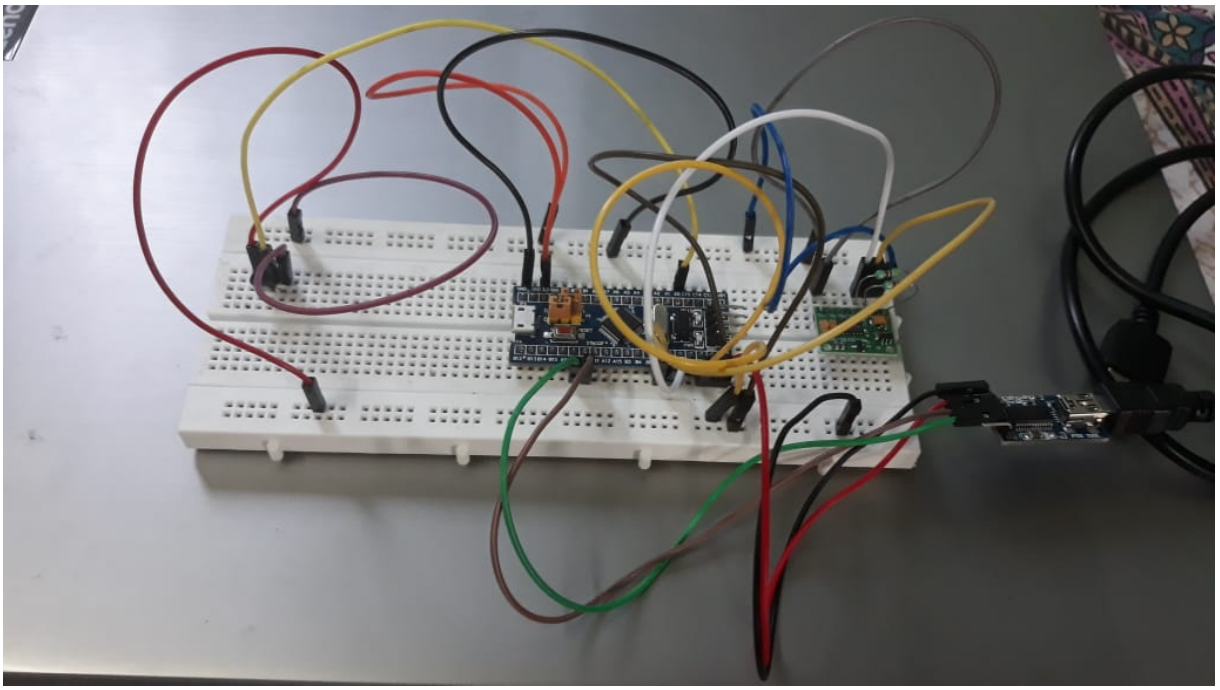


Figure 12. Actual Circuit

OUTPUT:

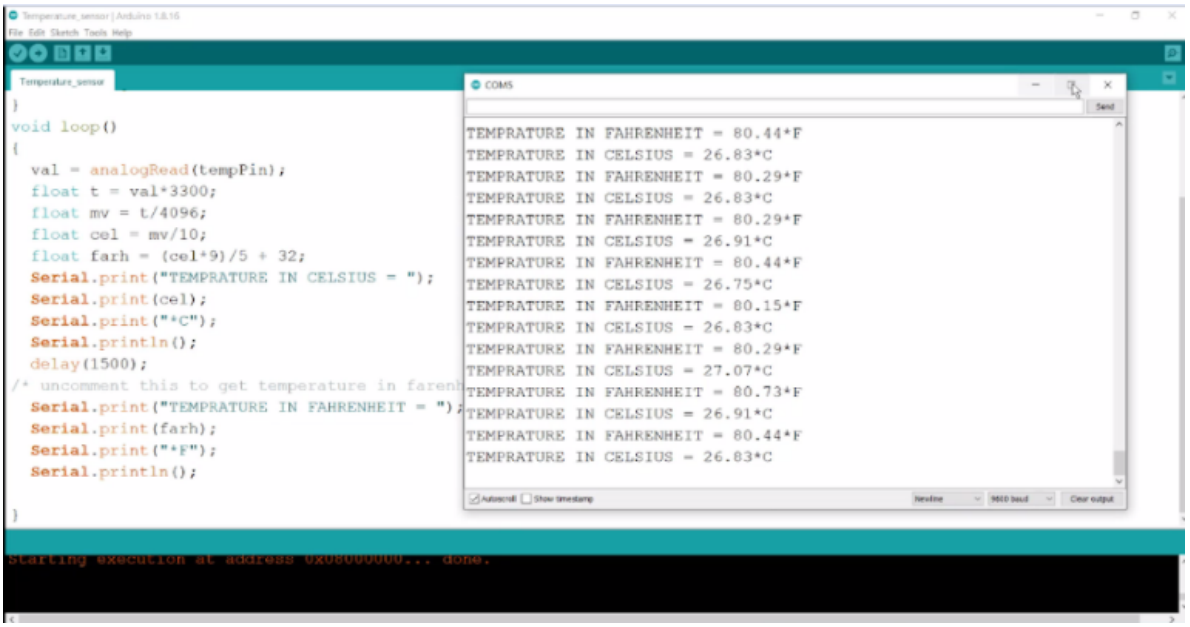


Figure 13. LM35 Reading

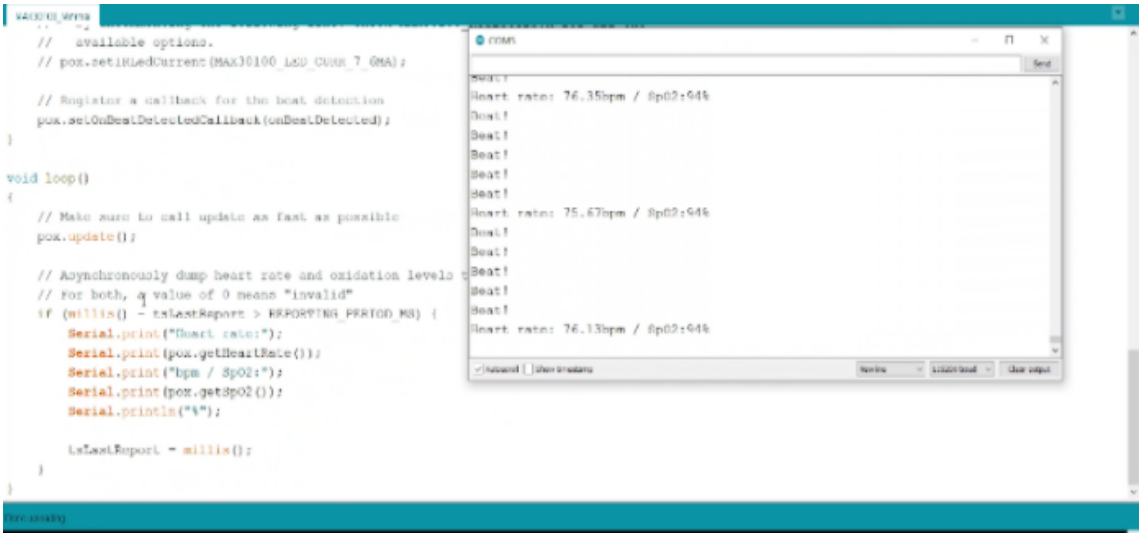


Figure 14. Oximeter reading

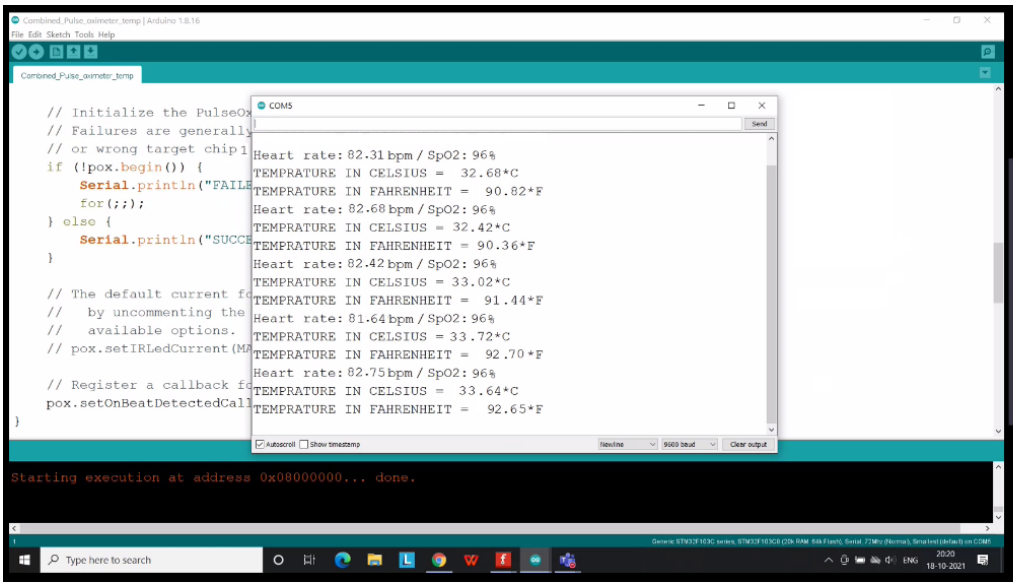


Figure 15. Interfacing temperature sensor and oximeter with stm32

CHAPTER 7:
APPLICATIONS

The STM32duino is a very good alternative to a regular Arduino such as the Uno, provided physical compatibility with existing Arduino shields is not required. It is much more powerful and costs a lot less.

Applications of STM32 Blue Pill

- Rapid prototyping
- CRC calculations
- Robotics
- Consumer products
- Drone controllers



